ECS 240 Project Proposal

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1 Introduction and Motivation

1.1 Problem Domain and Motivation

Functional programming allows for greater modularity.[1] Side effects cause difficulty in reasoning about code; unclear when calling a function which components are modified.[2] Yet effects are often absolutely necessary as with file I/O. Desire to break free from assignment and mutation[3]

1.2 Proposed Language

We propose a toy programming language similar to traditional systems programming languages like C and Rust with the distinction that every function is explicitly defined as "pure" or "impure", where a "pure" function may only modify state initialized in or moved into its scope. Our language will implement at least the minimum set of elements that allow us to illustrate this system, including variables, functions, references, and an 8-bit unsigned integer primitive as the sole type. We may support other features such as structs, strings, or simple file I/O depending on our timeline (see §§ 2 and 4).

Rust itself originally sought such a design, defining all functions as "pure" by default and employing the `io` keyword to explicitly designate a function with I/O side effects.[4] The language designers eventually opted to remove `io` keyword and the pure/impure separation. We hope to discover while designing and analyzing our own language what explicit declaration of side effects could enable for language users and whether the Rust developers' decision was ultimately justified.

```
fn pure_function(x, y) {
    x + y
}

// global variable
var z;

// You must specify the data out of scope whose
// state may change
fn impure(z) impure_function(x, y) {
    z = x + y;
}

// If variables have fields, those fields
// must be declared
fn impure(x.value, y.value) impure_function_struct(x, y) {
    // mutates x
    x.value = 1;
    y.value = 1;
}

// You can look at the program see a full list
// of all the things that could change state
fn impure main() {
    var x = 1;
    var y = 2;
    x = pure_function(x, y);
   // mutated: z
   impure_function(x, y);
}
```

1.3 Proposed Tool

compiler, where the input is a string in our source language and the output is a string in Rust. The user interacts with it from the command line As a stretch goal (see §2), an analysis tool to glean information from impure declarations may be in scope.

1.4 Scope and Related Work

Haskell has the Monad[5] Functional programming allows for greater modularity.[1] And they also often discourage side effects by the nature of their design. `cargo-scan` is a static analysis tool for Rust that detects side effects from Rust source code.[6]

2 Goals

Minimum goal. Parser, interpreter, and working compiler. Interpreter and compiler must detect whether a function is pure or impure and error if a function is impure without declaring so along with the outside state it modifies.

Target goal. The above, along with a static analysis tool that enumerates all the possible state that is modified given a function (including the functions it itself calls). The tool will at least be a CLI, but could be extended to a VSCode extension as well.

Reach goal. The above, as well as other Rust features that might make the division harder and mitigation: system state (file I/O), structs, etc.? Specifically for file I/O, we could explore options to avoid Rust's issue of `io` keyword explosion (similar to Haskell's Monad).

3 Syntax and Semantics

```
example program goes here
```

$$e ::= n \mid x \mid e_1 + e_2$$
$$c ::= x := e \mid c_1; c_2$$

4 Implementation Plan

4.1 Tool and Language Selection

Parser in Rust; meta language is Rust; target language for compiler is Rust.

4.2 Incorporating Existing Tools

From scratch.

4.3 Architecture

Core language definition; parser that outputs core; interpreter reads core and executes; compiler reads core and outputs IR, in IR we do checks on purity, then IR to target Rust

4.4 Development Effort

4.5 Challenges

Tracking scope is hard; tool is too much work... Fallback..? What if a variable will *never* change state? if we have if statements...should we detect that? need it be declared?

5 Collaboration Plan

Division of Work. Work divided as follows

Component	**Group Member(s)**
Core	Jacob Hamzawi
Parser	?
Interpreter	?
Compiler	?
Analysis tool	?

Communication. Slack and GitHub; Meetings after class on Monday and over Zoom on Thursday

Timeline. The below timeline assumes a 4-week time allotment to complete the project. It may be contracted or expanded depending on our eventual project due date.

Week	Components
Week 1	Parser
Week 2	Interpreter + Compiler
Week 3	Analysis tool + exploration
Week 4	Presentation

Commitment statement.

6 Conclusion and Questions for Feedback

Re-state goal kinda: does explicit knowledge of impure functions change programming paradigm? Or something like that.

Questions: are variables enough? Should we focus instead on system state? That's more difficult but perhaps more compelling.

References Cited

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